NX **NEXUS**
PQS HEALTH MONITOR · V3

PLATFORM
- Dashboard
- Product Network
- Signal Library 194
- Data Ingestion

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CONFIGURATION
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Meridian Pharma Ltd
ICH Q10 · EU GMP · 21 CFR 211

PQS Health Dashboard

Two-layer model · Layer A Enterprise / Layer B Product Network · March 2026

9 signals triggered | FI: High 61 | RI: Moderate 68 | GAMP Cat 3 ✓ | KF

SYSTEM STABILITY INDEX	FRAGILITY INDEX	PLCI · PRODUCT A	PLCI · PRODUCT B	RESILIENCE INDEX
72	61	75	64	68
Developing — Controlled	High fragility	Controlled · 5-node network	Drifting · 3-node network	Moderate
▼ −3 vs Feb	▲ +8 vs Feb	▲ +1 vs Feb	▼ −5 vs Feb	▼ −4 vs Feb

PQS Health Foundation

Four-index model · SSI · PLCI · Fragility Index · Resilience Index

SYSTEM STABILITY INDEX LAYER A
72
Developing · Controlled
▼ −3 MoM · Level 3/5
PQS structural stability — all 5 dimensions

PLCI — PRODUCT NETWORK LAYER B
75
Controlled · Product A 5-node
Prod A 75 · Prod B 64
Per-node · criticality-weighted · QTA governance

FRAGILITY INDEX A+B
61
High · structural weakness
▲ +8 MoM · watch
Where will the system break next?

RESILIENCE INDEX LAYER A
68
Moderate absorption capacity
▼ −4 MoM · supply risk
Can the PQS absorb disruption without losing control?

Layer A = Enterprise PQS · Layer B = Product Network (node-weighted, QTA governance) · Ref: NEXUS MVP Spec v3.0

NEXUS Stability Matrix

Fragility Index vs Resilience Index · current position marked ■

FRAGILITY INDEX →

	HIGH RESILIENCE	LOW RESILIENCE
	Robust System — Low risk · inspection ready · PQS absorbs stress	**Hidden Risk** — Stable today · not resilient to disruption
	Recoverable — Fragile but capable of recovery · watch	**Critical Risk** — Fragile + unable to absorb stress · act now

CURRENT
FI: 61 High
RI: 68 Mod
Position: critical-recoverable boundary. Priority: reduce FI by addressing CAPA backlog.
▲ Trending toward Critical

Five Dimension Scores

ICH Q10 / EU GMP weighted · 194-signal engine

GMP Process Control	74
Investigation & CAPA	65
Product Lifecycle	79
Supplier & Network	68
PQS Governance	76

Reactive — ▲ Level 3 — Controlled — Predictive

Top Fragility Index Drivers

Signals contributing most to FI 61 (High)

2013 **CAPA backlog growth** L1 Layer A
24% open CAPAs overdue vs 20% threshold. 3-month consecutive growth. Feeds RI System Capacity (62).

2037 **Supply chain dependency** L2 Layer B
74% single-source on 2 critical materials. Product A network: N1 CMO A single source. Feeds RI Supply Redundancy (58).

2043 **Emergency change rate** L2 Layer A
14% emergency vs 10% threshold. Governance Adaptability signal. Rising 60-day trend.

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NEXUS PQS HEALTH MONITORING · ICH Q10 §4.1 · 194-SIGNAL ENGINE · MERIDIAN PHARMA LTD

Executive PQS Health Reports

Data completeness 92% · 181 / 194 signals active · Confidence: High · QP: Dr K. Fitzgerald · GAMP Cat 3

72
SSI · DEVELOPING · ▼ -3 MoM

Monthly Executive Report · March 2026 | Annual Executive Dashboard · 2025–2026

SYSTEM STABILITY INDEX	PLCI · PORTFOLIO AVG	FRAGILITY INDEX	RESILIENCE INDEX	INSPECTION RISK SCORE
72	74	61	68	42
Developing · Level 3/5	Prod A 78 · Prod B 64	High · CAPA + supply risk	Moderate · supply + capacity	Elevated · ▲ +3 MoM

PQS MATURITY ASSESSMENT

Controlled Level 3 of 5 · SSI 65–79

PQS stable but primarily reactive. Investigation & CAPA dimension (65) is the primary lever. System capacity strained — investigation workload pressure feeds both the Fragility Index (61) and Resilience Index depression (68).

Gap to Proactive: +8 SSI points · Lever: Investigation & CAPA (65 → 75) + supply redundancy (>1 approved source)

NEXUS STABILITY MATRIX · MARCH 2026

Robust System FI Low · RI High	Hidden Risk FI Low · RI Low
Recoverable FI High · RI High	Critical Risk ● FI 61 · RI 68 · current position

▲ Trending toward Critical · FI growing +8/month · priority: reduce FI below 55 within 60 days

SIGNAL ALERT REGISTER — MARCH 2026 · 9 SIGNALS TRIGGERED

ID	SIGNAL	INDEX	METRIC	LAYER	REG. REF	RECOMMENDED ACTION
2013	CAPA backlog growth	A SSI·FI·RI	24% overdue	L1	Ch.1/8	QP monthly CAPA review · target +5 closures/month · prioritise oldest 5 by RCA category
2037	Supply chain dependency	B FI·RI:Sup	74% single-source	L2	Ch.7	Qualify alternate supplier for 2 critical materials. Target <70% by Q3 2026.
2043	Emergency change rate	A FI·RI:Gov	14%	L2	Ch.1	Root cause emergency change triggers · improve change forecast · target <10% April
2020	Investigation workload	A RI:Cap	>8 per FTE	L1	Ch.1/8	Assess QA resourcing. Triage open investigations. Escalate capacity risk to QP and CFO.
2034	Repeat supplier findings	B FI·SSI	2 repeat themes	L2	Ch.7	Expedite Precision Parts CAPA. Joint quality review within 30 days.

INSPECTION READINESS SCORE · MARCH 2026

INSPECTION RISK — 6-MONTH TREND & ACTIONS

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SSI · DEVELOPING · ▼ −3 MoM

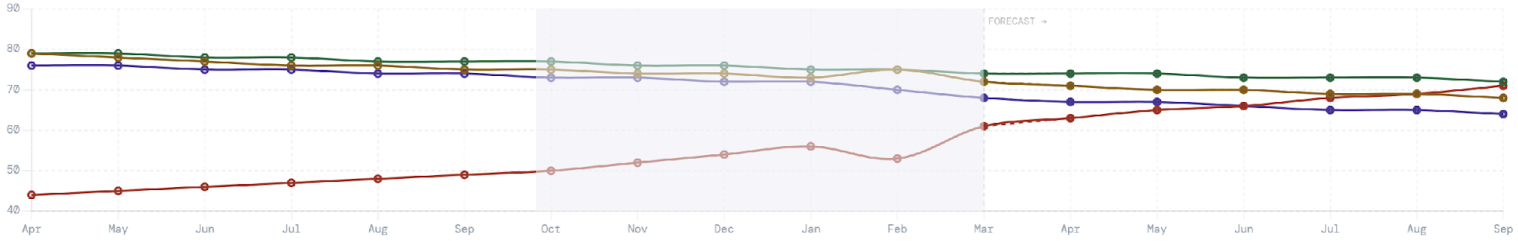
Monthly Executive Report · March 2026 | Annual Executive Dashboard · 2025–2026

SSI · 12-MONTH AVG	PLCI · 12-MONTH AVG	FRAGILITY INDEX · AVG	RESILIENCE INDEX · AVG
74	76	52	72
Peak 79 · Low 71 · Mar 72	Prod A stable · Prod B declining	Growing from 44 to 61 · trend ▲	Declining from 76 to 68 · trend ▼

Historical Trend — Four Indices · Apr 2025 – Mar 2026

12-month historical · 6-month predictive lookout · shaded band = forecast range

SSI | PLCI | Fragility Index | Resilience Index | Forecast (Apr–Sep 2026)



FORECAST Apr–Sep 2026 (±5 confidence band): SSI 69–73 (stable-declining without CAPA action) · FI 62–71 (worsening if no supply/CAPA action) · RI 65–70 (moderate decline) · PLCI 73–77 (stable)

Inspection Readiness Score — Annual View · 2025–2026

Based on SSI · PLCI · Fragility Index · Resilience Index · calculated monthly · Apr 2025 – Mar 2026

CURRENT · MARCH 2026

42
ELEVATED RISK

Annual trajectory: Low (24, Apr 2025) → Elevated (42, Mar 2026). Inspection observations likely if a regulatory visit occurs in the current window.

0–20 Low | 20–40 Moderate | 40–60 Elevated ◄ | 60–80 High | 80–100 Critical

SSI · avg	PLCI · avg	FI · avg	RI · avg
74	76		

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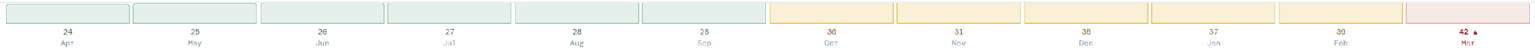

Annual avg IRS	Year end IRS	Annual change	Sep 2026 forecast
31	42	+18	~51
Moderate — year average	Elevated · crossed Dec 25	24 → 42 · worsening	No-action scenario

Audit Readiness Assessment — March 2026

Based on 12-month trend · signal register · index trajectories · ICH Q10 / EU GMP

HIGH RISK — CAPA Backlog — Investigation & CAPA
24% overdue CAPAs is a primary inspection finding. Growing trend over 3 months. Auditors will scrutinise CAPA effectiveness and recurrence rate. Resolve minimum 8 oldest CAPAs before audit window.

HIGH RISK — Supply Chain Single-Source Dependency
74% single-source dependency on 2 critical materials. EU GMP Ch.7 requires adequate supplier qualification. Demonstrate alternate supplier assessment or risk register mitigation for EU GMP audit readiness.

WATCH — Emergency Change Frequency — Governance
14% emergency changes signals reactive governance. Inspectors view emergency change rate as a governance maturity indicator. Document root cause analysis and corrective programme before inspection.

WATCH — Management Review Cadence (182 days)
ICH Q10 §4.1 requires periodic management review. 182 days exceeds recommended 180-day cadence. Schedule April review now. Ensure QP, CEO, and production lead attendance documented.

WATCH — Resilience Index Decline (76 → 68)
Declining 8 points in 12 months signals eroding system capacity. Regulators increasingly focus on systemic quality capability. Prepare narrative for inspection on resourcing and capacity management plans.

READY — Product Lifecycle Control · Product A
PLCI 78 and stable. APQR completion 95%. Stability data complete 100%. Product A supply chain well-documented. No action required — maintain current cadence.

6-MONTH PREDICTIVE LOOKOUT · APR – SEP 2026 (NO-ACTION SCENARIO)

INDEX	APR	MAY	JUN	JUL	AUG	SEP	SCENARIO	RISK
SSI	71	70	70	69	69	68	Slow decline if CAPA backlog not resolved	MEDIUM
Fragility Index	63	65	66	68	69	71	Worsening toward Critical if supply dependency + CAPA not addressed	HIGH
Resilience Index	67	67	66	65	65	64	Gradual decline as supply redundancy stays unresolved	MEDIUM
PLCI	74	74	73	73	73	72	Slight decline driven by Prod B (64 → 62). Prod A stable.	LOW

Forecast uses 6-month rolling signal momentum + linear regression on 12-month history. Confidence band ±5 points. With-action scenario available after baseline data complete (9/9 datasets).